



Assignment of master's thesis

Title: GPU implementation of acceleration data structures for ray tracing
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Study program: Informatics
Branch / specialization: Computer Systems and Networks
Department: Department of Computer Systems
Validity: until the end of summer semester 2026/2027

Instructions

- 1) Study the ray tracing (RT) rendering method and the acceleration data structures (ADS) used for it.
- 2) Analyze existing tools for visualization, testing, and debugging of ADS development and for performance evaluation of ADS computations during RT.
- 3) Study from [2] suitable parameters for testing functionality during the development and debugging of parallel GPU implementations of the ADSs and parameters for the performance evaluation.
- 4) Design and perform a GPU implementation of an ADS used in state-of-the-art RT [3]. Using the metrics from point 3), analyze the complexities of ADS construction and of scene traversal within RT. Compare your results with those in [3].
- 5) Extend the GPU implementation of the ADS from point 4) with Extended Morton codes [4] and Skewed Oriented Bounding Boxes [5]. Evaluate the individual extensions using the parameters from point 3) and analyze the complexities of ADS construction and of scene traversal within RT. Discuss the impacts of these 2 extensions.

Suggested literature:

- [1] Meister, Daniel, et al. "A survey on bounding volume hierarchies for ray tracing." Computer Graphics Forum. Vol. 40. No. 2. 2021.
- [2] Meister, Daniel, and Jiří Bittner. "Performance comparison of bounding volume hierarchies for GPU ray tracing." Journal of Computer Graphics Techniques (JCGT) 11.4 (2022): 1-19.



- [3] Meister, Daniel, and Jiří Bittner. "Parallel locally-ordered clustering for bounding volume hierarchy construction." IEEE Transactions on Visualization and Computer Graphics 24.3 (2017): 1345-1353.
- [4] Vinkler, Marek, Jiří Bittner, and Vlastimil Havran. "Extended Morton codes for high performance bounding volume hierarchy construction." Proceedings of High Performance Graphics. 2017. 1-8.
- [5] Káčerik, M., and Jiří Bittner. "SOBB: Skewed Oriented Bounding Boxes for Ray Tracing." Computer Graphics Forum. 2025.

